GREG FURLICH

Research Associate

greg.furlich@colorado.edu | 303-492-0021 | Boulder, CO

gregfurlich.com | nsi/greg-furlich | linkedin.com/in/greg-furlich | github.com/gfurlich

Research Faculty and the Space Domain Awareness research lead for the Center for National Security Initiatives at the University of Colorado Boulder. Leads the Data Exploitation Lab for Trusted Autonomy (DELTA) research group. Research interests include sensors, and sensing, mission data processing, and trusted autonomy. PhD in Physics from the University of Utah with a thesis on the ultraviolet remote sensing of ultra-high energy cosmic ray interactions within the atmosphere. Raised over $2.51 M in awards in the past 4 years and conferred an affiliation with the Aerospace Engineering Sciences at CU Boulder.

EDUCATION

Doctorate of Philosophy, *Physics* 2014 – 2020

Master of Science, *Physics* 2014 – 2018

University of Utah, Salt Lake City, UT

Bachelor of Science, *Physics* 2010 – 2014

Michigan Technological University, Houghton, MI *Magna Cum Laude*

Minors: Mathematical Sciences and German

RESEARCH INTERESTS

Sensors and Sensing: image collection, processing, and analysis over many spectral regimes (visible, infrared, ultraviolet, multispectral, hyperspectral); expeditionary low SWAP sensor systems for situational awareness; target signature phenomenologies

Mission Data Processing: signal, image, and video processing; feature extraction and exploitation; algorithm development for dim object detection and disparate data fusion for object tracking and state estimation; event detection, characterization, and typing; machine learning for automatic target recognition, stochastic processing modeling, anomaly detection, physics-informed neural networks

Trusted Autonomy: autonomous decisions for proliferated, distributed, and decentralized systems; battle management architectures; killchain or threat decomposition; course of action recommendation; hostility and intent assessment; risk assessment and management

RESEARCH EXPERIENCE

Research Associate Aug 2022 – Present

Space Domain Awareness Research Lead

University of Colorado Boulder,
Center for National Security Initiatives (NSI),
Data Exploitation Lab for Trusted Autonomy (DELTA)

- Leading the Data Exploitation Lab for Trusted Autonomy (DELTA) research group, a multidisciplanary group focused on applied physics, mathematics, and computer science for mission data processing for SDA, MWD, and ISR.
- Led many multi-disciplanary DoW and IC proposal efforts as CU Principal Investigator. Awarded $2.51 M in funding efforts and proposals.
- Led applied research projects in the USSF Space Systems Command (SSC) Boulder and Space Domain Awareness (SDA) Tools, Applications, and Processing (TAP) Labs. Selected for the inaugural Project Apollo Technology Accelerator cohort of the SDA TAP Lab and continued participation up to Cohort 9 before the SDA TAP Lab became the BMC3I TAP Lab.
- Developed and successfully tested a unrestricted operational capability for detection of launchs using publicly avaible weather satellite imagery. Capability is named Open-data Weather imagery Launch Alerts and Tracklets (OWLAT).

RESEARCH EXPERIENCE CONTINUED

- Briefed SDA capabilities and research impact to the then Chief of Space Operations (CSO), General Saltzman, then Vice Chief of Space Operations (VCSO), General Guetlein, Commander of Space Systems Command (SSC), General Garrant, then US Space Force - Space (S4S) Commander, General Schiess, Commander of Space Operations Command, General Miller, staffers from two US Senate Offices, and Congressional representatives from the House Armed Services Committee.
- Selected as a fellow for the AFOSR Summer Faculty Fellowship Program and invited to perform impactful research at the AFRL Information Directorate (AFRL/RI).

AWARDS

Advanced Maui Optical and Space Surveillance Technology Conference (AMOS) Best Poster	Sept 2025
Air Force Office of Scientific Research (AFOSR) Summer Faculty Fellow, AFRL/RI	Summer 2023
Recognized Technical Talent, Lockheed Martin	Selected 2021
Departmental Scholar, Department of Physics, Michigan Technological University	2013
Sigma Pi Sigma, Physics Honor Society	Inducted 2013
Michigan Space Grant Consortium Recipient	2012

SELECTED PUBLICATIONS AND PROCEEDINGS

Reinforcement Learning Agent Based Multi-Target Tracking with Closely Spaced Objects (CSOs), IEEE Aeroconf 2027, Big Sky, MT, *Abstract Accepted*

Transformer-Based Resident Space Object (RSO) Characterization from Photometric Time Series Observations, 2026 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS), *Accepted for Oral Presentation*

A Spatial-Spectral Convolutional Neural Network for Dim, Unresolved Target Detection in Multispectral Imagery, S Rossland, **G Furlich**, T Gill, B Sweeney, SPIE Defense + Security, 2026

Data-driven chromaticity-space rotations for enhanced target detection in multispectral overhead imagery, S Rossland, **G Furlich**, M W Chan, S Kerner, SPIE Defense + Security, 2026

Utilizing Civilian Launches as Live Exercises for Evaluating a Federated Protect and Defend SDA Battle Management System, **G. Furlich** et al., 2025 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS) (**Best Poster Award**)

Automated, Collaborative Applications to Close Kill Chain Gaps, **G. Furlich** et al., 2024 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS)

SubPixel Localization of Objects Using Multiple Spectral Bands, M. Gupta, J. Chan, M. Krouss, **G. Furlich**, P. Martens, M. Chan, M. L. Comer, E. J. Delp, IEEE Aerospace Conference, 2022

Infrared Small Target Detection Enhancement Using a Lightweight Convolutional Neural Network, M. Gupta, J. Chan, M. Krouss, **G. Furlich**, P. Martens, M. Chan, M. L. Comer, E. J. Delp, Geoscience and Remote Sensing Letters, 2022, doi: 10.1109/LGRS.2022.3203931.

SELECTED AWARDED GRANTS AND FUNDING CONTRACTS

Received **21 Awards**: **$ 2.51 M total** funding, **11 unique** awards, **10 extensions**, with awards from over **11 sponsors**.

SDA TAP Lab Cohort Awards, Institutional PI, **$265,000** (with multiple extensions) — Jun 2024 – Mar 2026
Open-Source Launch Detection

Aerospace Corporation Award, PI, **$270,000** (with multiple extensions) — Sept 2024 – Aug 2027
University Partnership Program: SDA TAP Lab Research Projects

Lockheed Martin Space Systems IRAD, PI, **$454,000** (with multiple extensions) — Jan 2024 – Aug 2026
Multi-spectral Dim Object Detection and Multi-Object Tracking

NSTXL OTA # FA8814-21-9-0001, Institutional PI (Prime: Lockheed Martin Space Systems), **$968,000** — Apr 2023 – Mar 2027
Overhead Persistent Infrared Data Exploitation Technology Transition, OPIR ODETT

AFOSR Summer Faculty Fellowship Program, PI, **$24,240** — June – Sept 2023
Orbit-to-tracklet Hypothesis Constraints to Frame Shifting in a Track-Before-Detect Method for Wide Field of View Survey Telescopes